LIANGZHENG (DAVID) LIU

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Engineering Portfolio: https://box58m.github.io/index.html

EDUCATION

University of Illinois at Urbana-Champaign — Expected May 2027
Bachelor of Science in Systems Engineering and Design (SFO in Autonomous Systems and Robotics)
Minor in Computer Science

RELATED COURSEWORK

CAD, Engineering Materials, Data Analysis and Visualization, Dynamic Analysis, Fluid Mechanics, Structure/Mechanism Design and Analysis, Optimization, Closed Loop Control Systems, State Space Design for Control, Introductory Circuitry

TECHNICAL SKILLS

Programming Languages: C++, Python, Structured Text, Java
Engineering/Design: Fusion 360, MATLAB, CODESYS, Factory I/O
Fabrication: Laser Cutter, Additive Manufacturing, Drill Press, Band Saw

ENGINEERING PROJECT HIGHLIGHTS

PLC-Controlled Automated Manufacturing and Storage System

CODESYS, Factory I/O, PLC Automation, Structured Text

- Programmed a multi-station production line integrating material feeding, fabrication, buffering, dispatch, robotic assembly, inspection, pallet handling, and automated vertical storage.
- Developed modular state-machine and function-block logic with sensor interlocks, synchronized material release, age-priority dispatch, and timeout-based fault recovery.
- Documented the system architecture, operating sequence, control logic, demonstrations, and complete CODESYS source code in a deployed engineering portfolio.

Self-Balancing Robot Controller

C++, Arduino, LSM6 IMU, Quadrature Encoders

- Implemented real-time balance control using fused angle estimation and proportional, angular-rate, wheel-velocity, and wheel-position feedback.
- Improved physical performance through sensor-offset calibration, feedback-sign validation, motor dead-zone compensation, command limiting, leakage, and left-right motor synchronization.
- Designed a button-triggered kick-and-brake state machine that raises the robot from rest, transitions into balancing, and rejects external disturbances.

Mobile Gantry Crane Prototype

Systems Design, Component Design, Fusion 360, Arduino

- Applied a V-model systems-engineering process to translate functional requirements for base travel, carriage motion, and vertical lifting into subsystem designs and corresponding validation tests.
- Evaluated design constraints—including available materials, actuation hardware, fabrication capabilities, and project schedule—to guide architecture and component-selection decisions.
- Designed and fabricated an extrusion-guided carriage, custom winch and pulley components, and a four-bar drive suspension; integrated Arduino-controlled DC motors and validated all primary system functions.

WORK EXPERIENCE

UIUC Engineering Systems Design Laboratory — Champaign, IL

Laboratory Assistant — August 2026 – Present

- Calibrate and verify laboratory equipment to support reliable operation for engineering instruction and project work.
- Fabricate and modify mechanical components using additive manufacturing, laser cutting, and general shop tools.
- Assist with equipment setup, maintenance, and troubleshooting for engineering laboratory activities.

UIUC Engineering Services Surplus — Champaign, IL

Inventory Management — May 2024 – August 2025

- Supported semiannual equipment audits by tracking asset locations and verifying functionality for 400+ scientific instruments and 200+ computing devices.
- Developed an item taxonomy and location schema to improve asset organization, traceability, and audit consistency.
- Redesigned warehouse storage using 5S principles and visual management, increasing capacity by more than 20% and reducing retrieval time for frequently handled equipment.
- Standardized the processing workflow for surplus computing equipment, reducing transportation and over-handling waste while improving processing speed by 30%.

Xi'an Tian-Dun Cylinder Gasket Co. — Remote

Elastomer Qualification Program Coordinator — September 2025 – Present

- Coordinate communication among suppliers and Cummins stakeholders to support elastomer qualification milestones and technical decisions.
- Translate and verify technical documents covering test procedures, standards, and results for cross-functional communication.
- Track test progress, qualification records, documentation, and results to maintain visibility across the program.

MATERIAL TESTING LABORATORY

- Performed laboratory experiments on material strength, hardness, ductility, and failure characteristics under controlled loading conditions.
- Analyzed stress-strain data and compared failure modes across engineering materials.
- Interpreted test results to relate material properties to engineering design considerations.
- Produced technical reports with tables, visual summaries, and stress-strain curves to present findings clearly and support engineering conclusions.

EXTRACURRICULAR ACTIVITIES & LEADERSHIP

Illini VEX Robotics — Champaign, IL

Design Team / Event Organizer — December 2024–Present

- Accelerated development cycle by implementing rapid prototyping techniques such as additive manufacturing and laser cutting.
- Designed robot components in Fusion 360 according to team requirements.
- Worked closely with the executive team to coordinate event logistics.

CERTIFICATIONS

- Six Sigma Green Belt certification by IISE. — November 2025
- Lean Green Belt certification by IISE. — March 2026